Computational PDEs — Day 4 classwork

Exercises (about 45–60 minutes)

Hiroyuki Miyoshi
The University of Tokyo / ELTE Erasmus+ ICM

12 November 2026

Format

Two questions: Q1 is by hand; Q2 uses the notebook `sc_prime.ipynb`.
Work in pairs if you like.

1. **Möbius map** — unit disk to the right half-plane.
2. **Schottky–Klein map** — run `sc_prime.ipynb` and vary the domain.

1 Question 1 — Unit disk to the right half-plane

On the simply connected case in class, the map is

$$g(\zeta) = -\frac{\zeta - 1}{\zeta + 1}, \qquad |\zeta| < 1. \tag{1}$$

Goal. Show by hand that g sends the unit disk $|\zeta| < 1$ into the right half-plane $\{\, w : \Re w > 0 \,\}$, with the unit circle mapped to the imaginary axis $\{\, \Re w = 0 \,\}$ (apart from the pole at $\zeta = -1$ on the circle).

1. **The centre.** The origin $\zeta = 0$ lies in the disk. Compute $g(0)$. What does this suggest about the image of the interior?
2. **Two test points.** Find $g(\mathrm{i})$ and $g(1)$. (For $\zeta = 1$ you are on the boundary $|\zeta| = 1$; the value is still defined by (1).)
3. **The unit circle.** Let $|\zeta| = 1$ and $\zeta \neq -1$. Write $\zeta = e^{\mathrm{i}\theta}$ with $\theta \in (-\pi, \pi) \setminus \{\pi\}$. Show that

$$g(\zeta) = -\mathrm{i} \tan \frac{\theta}{2}, \tag{2}$$

   so $\Re g(\zeta) = 0$. Conclude that the boundary is mapped onto the imaginary axis (a vertical line through 0).

Hint for Q1 part (3)

Factor $e^{i\theta/2}$ from $\zeta \pm 1$:

$$e^{i\theta} - 1 = 2i\, e^{i\theta/2} \sin(\theta/2),$$
$$e^{i\theta} + 1 = 2\, e^{i\theta/2} \cos(\theta/2).$$

Q1 (1) — $g(0) =$ _____ interpretation: _____

Q1 (2)

$g(i) =$ _____ $g(1) =$ _____

Q1 (3) — short derivation ($\Re g = 0$ on $|\zeta| = 1$):

---

2 Question 2 — Multiply connected map in `sc_prime.ipynb`

Open and run `sc_prime.ipynb` from the course GitHub materials (Schottky–Klein prime calculator). After the first cells, use the Day 4 block at the end (domain / image panels).

The **baseline** geometry in the notebook is two holes:

$$qv = [0.1,\ 0.1], \qquad dv = [0.4,\ -0.4],$$

with `level = 5` when building `maps`. The conformal map to plot is

$$g(\zeta) = -\frac{\omega(\zeta, 1)}{\omega(\zeta, -1)}. \tag{3}$$

Each time you change `qv`, `dv`, or `level`, re-run the **first cell** (so `maps` is rebuilt) before the plotting cells.

1. **Larger holes.** Increase both radii in `qv` a little (for example try `qv = [0.15, 0.15]` or `[0.12, 0.14]`). Compare the image of g in the bottom-right panel with the baseline. If part of the boundary leaves the square, change `IMAGE_LIM` (default `(-2.0, 2.0)`) so the picture stays readable. *What changes in the w-plane?*
2. **Holes closer together.** Keep `qv` fixed and move the centres in `dv` closer (smaller distance $|dv[0] - dv[1]|$, still inside the unit disk and without overlap). Describe how the image of g changes, especially near the mapped hole boundaries. If the plot looks noisy or boundaries wiggle, raise `level` to `6` (or `7`) and re-run from the first cell.
3. **A third hole.** Add one more pair to `qv` and `dv` (now $M = 3$ holes, four boundary circles in total). Run the notebook and record what the image looks like. Adjust `IMAGE_LIM` and `level` if needed.

Q2 (1) — parameters tried and what you see:

qv=_____ IMAGE_LIM=_____

Q2 (2) — centres / level and observation:

dv=_____ level=_____

Q2 (3) — third hole (qv, dv) and observation:

What to watch

Use the **bottom row** (image) of the 2×3 figure: column (3) is (**??**). Thick black curve: outer circle $|\zeta| = 1$; coloured curves: hole boundaries.

Time box

About 45–60 minutes total. Q1 (1)–(2) first; then Q2 at a machine with the notebook kernel (`prime_mat` or equivalent).